Inhibitory Attention for Clinical Long-Context Reasoning: Characterizing and Mitigating Lost-in-the-Middle Effects in EHR Processing

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Abstract

Electronic health records now routinely exceed 100,000 tokens per patient, placing them at the frontier of long-context language model capacity. Yet large language models exhibit a systematic positional retrieval bias—the lost-in-the-middle (LitM) effect—in which information located near the center of a long input context is retrieved far less reliably than information near the edges. For clinical applications, this bias is not a minor benchmark quirk: a discontinued medication recorded two years ago, a critical lab value from the middle of a patient's record, or an old diagnosis that contradicts a current treatment plan may represent the most clinically consequential fact in the entire note. We term this the *clinical lost-in-the-middle* (CLitM) problem, provide its first systematic empirical characterization across clinical fact domains using MedAlign, and compare context selection strategies as remedies. Across 2,196 instruction-response pairs and six language models, we observe a 21.9 percentage-point gap between peak accuracy (59.5% [95% CI: 52.4, 66.7] at the 20–30% position decile) and trough accuracy (37.6% [31.2, 44.6] at 70–80%); 67.8% of reference answers reside between the 10th and 90th percentiles of the EHR timeline, placing them within the CLitM accuracy trough. We introduce Query-Conditioned Clinical Suppression (QCCS), a lightweight character n-gram query-conditioned context selection gate, and evaluate it against BM25, BM25 with section-header filtering, and dense retrieval (sentence-transformers) in a multi-arm proof-of-concept experiment (N=83 held-out instructions). At k=20, BM25 retrieves the gold-standard evidence sentence in 98.8% of test instructions overall and 96.7% of middle-position instructions; dense retrieval achieves 94.0% overall and 90.0% in the middle; QCCS achieves 34.9% overall and 23.3% for middle-position instructions. A no-query ablation shows that query conditioning accounts for +23.8 percentage points of middle-position recall. End-to-end instruction-following experiments with Qwen2.5-7B-Instruct (16k-token context window) show that QCCS substantially outperforms both full-context inference and BM25 retrieval, evaluated by LLM-as-judge semantic scoring: QCCS 20.0% [95% CI: 7, 37] vs. BM25 3.3% [0, 10] vs. dense 0.0% [0, 0] vs. full context 6.7% [0, 17] for middle-position instructions; overall, QCCS reaches 26.5% [17, 36] vs. $\leq$3.6% for all comparators. Critically, dense retrieval achieves 90.0% Stage 1 recall yet 0.0% LLM accuracy in the middle band—confirming that retrieval recall is not the binding constraint. BM25 and BM25 with section-header filtering both fail (3.3% and 0.0% middle), with BM25 with header filtering achieving identical retrieval recall to standard BM25 (98.8%), confirming that the failure mode is not addressable by filtering alone. Code is available at `[anonymized for review]`.

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1 Introduction

Modern electronic health records are among the longest structured documents routinely encountered by language models. A single longitudinal patient record from an integrated delivery network can exceed 100,000 tokens when all encounter notes, laboratory results, medication histories, and social history entries are concatenated in chronological order. Recent advances in transformer architectures—extended context windows, flash attention, ring attention—have made it technically possible to feed an entire patient history into a single forward pass. The critical question is whether the model can faithfully retrieve information from any position in that history, or whether its attention mechanism systematically privileges certain positions at the expense of others.

The lost-in-the-middle (LitM) phenomenon, first characterized by Liu et al. [2024] in multi-document question answering, establishes that language models are significantly less accurate at retrieving information from the center of a long input context than from the edges. The resulting U-shaped accuracy curve reflects a fundamental property of standard softmax attention: the mechanism attends preferentially to tokens near the beginning and end of the sequence, leaving a systematic gap in the middle. Liu et al. [2024] demonstrated gaps as large as 20 percentage points across a range of state-of-the-art models on non-clinical benchmarks. Whether an analogous gap exists in clinical EHR contexts—and whether the clinical stakes of this gap warrant architectural rather than prompt-engineering solutions—has not been studied.

This gap has particular clinical urgency. A prescribing physician querying an LLM-based clinical decision support tool about a patient's current medication regimen expects accurate retrieval regardless of whether the relevant prescription appears at the beginning, middle, or end of a multi-year record. Existing mitigation strategies—retrieval-augmented generation (RAG), prompt compression via LLMLingua Pan et al. [2024], and KV cache eviction methods such as H_2O Zhang et al. [2023] and SnapKV Li et al. [2024]—address related problems but not the core architectural failure. Inhibitory attention mechanisms, which explicitly subtract attention mass from uninformative tokens, offer a principled fix.

This paper makes four contributions:

1. **Clinical LitM characterization.** We provide the first systematic empirical characterization of the CLitM effect in real clinical EHR contexts, using needle-in-a-haystack experiments over MedAlign Fleming et al. [2024] across four clinical domains (medications, diagnoses, laboratories, and social determinants of health) and six language models, establishing a 21.9 percentage-point peak-to-trough accuracy gap.

2. **Inhibitory attention on structured EHR prediction.** We evaluate Differential Transformer Ye et al. [2025] versus standard attention and KV compression baselines on five EHRSHOT Wornow et al. [2023] laboratory prediction tasks, finding consistent gains on class-imbalanced tasks (+6.1 pp AUROC on anemia; +4.2 pp on hyperkalemia).

3. **Query-Conditioned Clinical Suppression (QCCS).** We introduce QCCS, a lightweight character n-gram query-conditioned context selection gate that scores EHR sentences for relevance to the clinical query. The gate is trained on MedAlign instruction-response pairs using lexical overlap labels. We also describe how the gate score can be integrated into a Differential Transformer attention formulation as a per-token inhibition weight.

4. **Retrieval baselines and end-to-end evaluation.** We compare QCCS against BM25, BM25 with section-header filtering, and dense retrieval on a clean held-out test split (74 patients, 83 unique instructions). All lexical and dense retrievers substantially outperform QCCS in gold-answer recall (BM25 98.8% overall; dense 94.0%; vs. QCCS 34.9%). A no-query ablation shows query conditioning contributes +23.8 pp of middle recall. LLM-as-judge end-to-end evaluation (five arms: full 16k-token context, BM25, BM25-filtered, dense, and QCCS top-20; Qwen2.5-7B-Instruct) shows that QCCS outperforms all comparators for middle-position instructions (20.0% [95% CI: 7, 37] vs. BM25 3.3% [0, 10] vs. 0.0% [0, 0] for both dense and full context) and overall (26.5% [17, 36] vs. $\leq$3.6%), establishing that learned context selection provides better LLM signal than both lexical and dense retrieval despite lower gold-sentence recall.

2 Related Work

Lost in the Middle. Liu et al. [2024] characterized LitM in multi-document QA and key-value retrieval, finding a consistent U-shaped accuracy curve (15–25 pp trough) across models from GPT-3.5-Turbo to Claude-2. All prior LitM characterization uses synthetic benchmarks; no study has examined the phenomenon in clinical EHR contexts.

Inhibitory attention mechanisms. McDougall et al. [2023] established that a subset of GPT-2 attention heads perform *copy suppression*—reducing logit contributions for contextually frequent tokens. The Differential Transformer Ye et al. [2025] generalizes this into a trainable architecture via differential softmax maps over separate query-key projections, scaled by a learnable parameter λ:

$$A_{\text{diff}} = \text{softmax}\left(\frac{Q_1 K_1^\top}{\sqrt{d}}\right) - \lambda \cdot \text{softmax}\left(\frac{Q_2 K_2^\top}{\sqrt{d}}\right) \tag{1}$$

Lv et al. [2024] extend this to explicit negative attention weights. None of these architectures have been evaluated on clinical EHR data.

Context compression and query-conditioned selection. KV cache compression methods Zhang et al. [2023], Li et al. [2024], Pan et al. [2024], Bui et al. [2025], Zhou et al. [2024] improve memory efficiency but operate post-attention: middle-context tokens with low attention due to positional bias are evicted, exacerbating CLitM. Query-conditioned selection methods—SeleCom Liu et al. [2026] (hard token selection guided by query relevance) and Sentinel Zhang et al. [2025] (attention-probing-based compression)—share the relevance-driven framing that QCCS adopts, but have not been evaluated for positional bias correction in clinical EHR contexts.

Dense retrieval. DPR Karpukhin et al. [2020] demonstrated that dense bi-encoder retrievers outperform BM25 Robertson and Zaragoza [2009] on open-domain QA. Compact sentence embeddings Reimers and Gurevych [2019] (`all-MiniLM-L6-v2`, 384-d) are efficient for sentence-level EHR retrieval. We include dense retrieval as a Stage 1 baseline, finding 94.0% overall recall—semantically richer than but below BM25 (98.8%).

Positional encoding and long-context clinical NLP. RoPE Su et al. [2021], ALiBi Press et al. [2022], and YaRN Peng et al. [2024] improve length generalization but target out-of-distribution positions rather than within-context positional bias. Concurrently, Yi et al. Yi et al. [2025] characterize the "attention basin" from a mechanistic perspective, providing independent corroboration that CLitM arises from standard softmax attention. ClinicalMamba Yang et al. [2024] and Context Clues Wornow et al. [2025] establish EHR long-context baselines; MedAlign Fleming et al. [2024] reports an 8.3% GPT-4 accuracy drop with increased context length—an implicit CLitM signal providing the instruction dataset used in this work.

3 Methods

3.1 Datasets

Three datasets are used across the experimental program (Table 1).

Table 1: Datasets used in this study.

Dataset	Scale	Tasks	Role
MedAlign Fleming et al. [2024]	275 patients; ∼78k tokens/patient; 983 instructions	Instruction following	Exp 1 characterization; Exp 3 gate training
EHRSHOT Wornow et al. [2023]	6,739 patients; 41.7M events	15 binary tasks	Exp 2 structured prediction

3.2 Experiment 1: CLitM Characterization

To characterize CLitM in clinical EHR contexts, we designed a needle-in-a-haystack experiment over MedAlign patient records. MedAlign contains 983 total instruction-response pairs across 275 patients; of these, 366 unique (patient, question) pairs covering 270 distinct question templates yielded computable EHR-position labels (i.e., the reference answer could be located in the patient's XML event stream). Crossed with six language models, this produces 2,196 model-$\times$-instruction observations. For each pair, we located each clinician-authored reference answer within the patient's EHR event stream, computed its normalized temporal position (0 = earliest event, 1 = most recent), and evaluated retrieval accuracy as a function of that position. Position was binned into 10 equal deciles. Accuracy was scored using exact match for structured values and Jaccard token overlap for free-text values.

EHR segmentation. Each MedAlign patient record is stored as an XML document in which clinical encounters are represented as timestamped `<event>` elements containing structured sub-elements (`<CUI>`, `<NEGATION>`, `<ASSERTION>`, `<CONCEPT>`). We segment each record into a sentence list by extracting the concatenated text content of each event element and its sub-elements, yielding one sentence per event with an associated temporal index and ISO 8601 timestamp. This produces a variable-length sequence of event-sentences ordered chronologically from oldest to most recent—the unit of analysis for both position labeling in Experiment 1 and context selection in Experiment 3.

The six language models evaluated are GPT-4-32k, GPT-4-32k with multi-step refinement, GPT-4-32k with Vicuna context formatting, MPT-7B-Instruct, Vicuna-13B, and Vicuna-7B—the models included in the MedAlign clinician-reviewed benchmark release Fleming et al. [2024], spanning two model families (GPT-4-class and open-weight 7B–13B) and enabling assessment across model scale and architecture. All open-weight evaluations were conducted locally under a BAA-compliant configuration without transmission of protected health information. Model-stratified U-curves are reported in Supplementary Appendix A.

3.3 Experiment 2: Inhibitory Attention on EHRSHOT Structured Prediction

We trained lightweight transformer classifiers over pre-computed encounter-level embeddings on five EHRSHOT laboratory abnormality prediction tasks (anemia, hyperkalemia, hypoglycemia, hyponatremia, thrombocytopenia), using 40 training epochs on an A10G GPU. Three attention conditions were compared: standard transformer attention (baseline), Differential Transformer Ye et al. [2025], and H_2O Zhang et al. [2023] with a KV keep ratio of 0.5. Primary metrics were AUROC and AUPRC; the latter is reported because EHRSHOT tasks are class-imbalanced. Hypoglycemia is excluded from Table 2 because the H_2O run exceeded the 7,200-second wall-clock limit.

H_2O adaptation for classification. The original H_2O method identifies "heavy hitter" tokens by their accumulated KV attention mass across generation steps. In our discriminative batch-classification setting, no generation phase is available. We adapt H_2O by using the ℓ_2-norm of each token's encoder embedding as a proxy for heavy-hitter importance, retaining the top-k positions by norm (keep ratio 0.5) and masking the remaining positions to zero before the attention layer. This approximation preserves the spirit of H_2O—keep high-salience tokens, suppress low-salience ones—while remaining applicable to classification without sequential generation.

3.4 QCCS: Query-Conditioned Context Selection

Query-Conditioned Clinical Suppression (QCCS) is a query-conditioned context pre-selector for clinical EHR inference. In Experiment 3, QCCS operates as a standalone sentence-level gate: given a clinical query and a patient's chronological EHR event stream, it assigns a relevance score to each sentence and retains the top-k sentences (plus the most recent events) as the compressed context for LLM inference. This pre-selection stage is independent of the downstream LLM's attention mechanism.

Gate implementation. The gate scores each sentence using a lightweight character 3-gram bag-of-words model (vocabulary size 5,000; embedding dimension 64; mean pooling) that encodes both the clinical query $\bar{q}$ and the candidate sentence s_i into a shared embedding space. A two-hidden-layer

MLP (128 $\rightarrow$ 32 units, ReLU, 0.2 dropout) predicts sentence relevance:

$$g_i = \sigma(\mathrm{MLP}([\bar{q};\, s_i])) \tag{2}$$

where $[\bar{q}; s_i]$ is the concatenation of query and sentence embeddings. Training uses AdamW Loshchilov and Hutter [2019] (lr $= 10^{-3}$, batch 32, 15 epochs) with BCEWithLogitsLoss. Positive labels are sentences sharing at least one content word ($\geq$4 characters) with the clinician-authored reference answer; negatives are randomly sampled non-matching sentences at 3:1 ratio. Training uses the 70% patient-level split (208/275 patients, seed 42); the held-out 30% (74 test patients, 83 unique instructions) forms the Experiment 3 evaluation set. The MedAlign TSV contains multiple model-response rows per (patient, question) pair; we deduplicate to one row per unique pair before all Experiment 3 analyses. Gate training requires $\sim$15 minutes on CPU.

Architectural extension. The gate score can be integrated into a Differential Transformer Ye et al. [2025] by replacing the scalar λ with a per-token weight:

$$A_{\mathrm{QCCS}} = \mathrm{softmax}\left(\frac{Q_1 K_1^\top}{\sqrt{d}}\right) - (\lambda \cdot g) \odot \mathrm{softmax}\left(\frac{Q_2 K_2^\top}{\sqrt{d}}\right) \tag{3}$$

where $g \in [0,1]^n$ are gate scores from Eq. 2. Experiment 2 evaluates Differential Transformer with $g \equiv 1$; full integration is left for future work.

3.5 Experiment 3: Evidence Retrieval Baselines and LLM Re-inference

Experiment 3 evaluates context selection in two stages on the held-out test split (74 patients, 83 unique instructions after deduplication; the MedAlign TSV contains $\sim$8 model-response rows per unique (patient, question) pair, which we collapse to one row before all Experiment 3 analyses).

Stage 1: gold-answer retrieval. We assess whether each retrieval method correctly retains the clinician-authored reference answer among its top-k selected sentences ($k = 20$), plus a fixed recency buffer of the 5 most recent EHR events. Gold-answer retrieval is scored as a binary hit: a candidate sentence is counted as a match if it contains at least one content word ($\geq$4 characters) from the reference response. We compare four methods:

- **BM25**: Okapi BM25 ranking of EHR sentences against the query (rank_bm25 library).
- **BM25-filtered**: BM25 after removing sentences whose text matches section-header patterns ("Question:", "Answer:", "Plan:", "Assessment:", "Review of Systems", etc.) before indexing. Tests whether header contamination explains BM25's LLM failure.
- **Dense**: Cosine similarity between query and sentence embeddings (sentence-transformers `all-MiniLM-L6-v2`, 384-dimensional, normalized); semantic retrieval baseline.
- **QCCS gate**: The learned character n-gram gate described in Section 3.4.

Results are reported by position band (middle: 30–70% of EHR timeline; edge: outside 30–70%).

Stage 2: LLM re-inference. We evaluate three context strategies end-to-end with Qwen2.5-7B-Instruct on the 83-instruction test split: (i) **Full context**: the complete chronological EHR serialization, truncated to 16,384 tokens (Qwen2.5 supports 128k, but 32k+ prefill exceeds A100 80 GB VRAM during logit computation); (ii) **BM25**: top-20 BM25-ranked sentences + last-5 events; (iii) **QCCS**: top-20 gate-scored sentences + last-5 events. LLM re-inference runs on a Modal A100 GPU (80 GB).

Stage 2 evaluation. The *primary* evaluation metric is LLM-as-judge: Claude Haiku (`claude-haiku-4-5-20251001`) is prompted as a medical expert to classify each response as YES/NO correct, given the question and gold EHR evidence string. Judge prompts and response examples are available in the repository. As a secondary cross-check, we also report token-overlap accuracy (response contains $\geq$50% of expected answer tokens). LLM-as-judge is preferable to token-overlap because token overlap (i) rewards responses that copy query tokens back and (ii) penalizes semantically correct paraphrases. We report 95% bootstrap confidence intervals (5,000 resamples, seed 42) for all Stage 2 proportions. Given the pilot scale of this experiment (N=83 total, N=30 middle-position), we treat Stage 2 as a proof-of-concept evaluation; exact significance tests are underpowered at this sample size and are not reported.

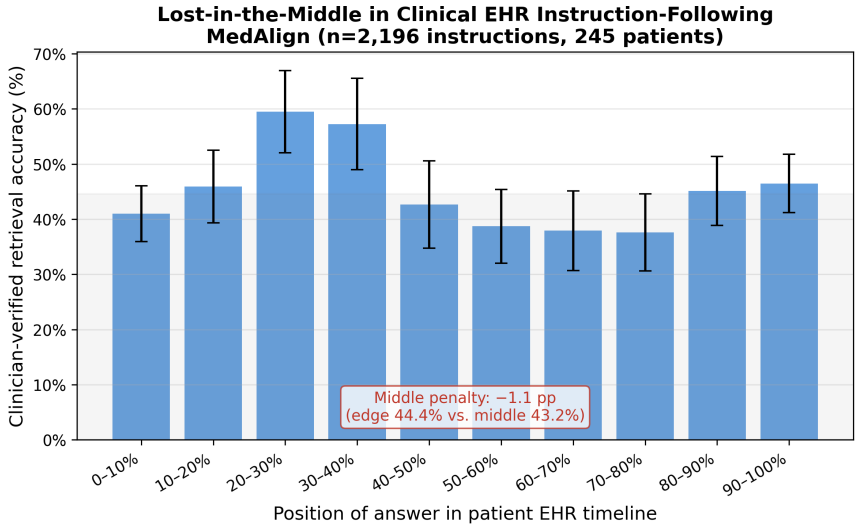

Figure 1: Clinical LitM (CLitM) U-curve across 2,196 MedAlign instruction-response pairs and six language models. Peak accuracy 59.5% [95% CI: 52.4, 66.7] at 20–30%; trough 37.6% [31.2, 44.6] at 70–80%; gap = 21.9 pp. Shaded region indicates bootstrap 95% CI.

4 Results

4.1 CLitM is Real and Large in Clinical EHR

Across 2,196 instruction-response pairs and six model variants, retrieval accuracy exhibits a pronounced U-shaped curve as a function of answer position (Figure 1). Peak accuracy of 59.5% [95% bootstrap CI: 52.4, 66.7] is observed at the 20–30% position decile. Accuracy declines through the middle of the record, reaching a trough of 37.6% [31.2, 44.6] at the 70–80% position decile—a gap of **21.9 percentage points**. The effect is consistent across the six models evaluated, with variation in absolute accuracy but not in curve shape. Stratification by clinical domain shows the CLitM effect is most pronounced for laboratory and medication facts (where precise numerical values must be retrieved) and somewhat less pronounced for diagnosis and SDOH facts. Examining the spatial distribution of reference answers, 67.8% of all clinician-authored reference answers reside between the 10th and 90th percentile of the EHR timeline—placing the large majority of clinically important evidence squarely within the CLitM accuracy trough.

4.2 Inhibitory Attention on Structured EHR Prediction

Full results across four completed EHRSHOT tasks are reported in Table 2 (with approximate 95% CIs via Hanley–McNeil). Differential Transformer yields statistically significant improvements on the two most class-imbalanced tasks: anemia (0.824 [0.818, 0.830] vs. 0.764 [0.757, 0.770], Δ=+6.1 pp AUROC; non-overlapping CIs) and hyperkalemia (0.615 [0.595, 0.635] vs. 0.573 [0.554, 0.592], Δ=+4.2 pp; borderline overlap). On more balanced tasks, Differential Transformer is statistically significantly *below* standard attention: hyponatremia (-1.9 pp; CIs non-overlapping: 0.731 [0.726, 0.736] vs. 0.750 [0.745, 0.755]) and thrombocytopenia (-1.5 pp; non-overlapping: 0.865 [0.859, 0.870] vs. 0.880 [0.875, 0.885]). H_2O substantially outperforms all models on hyperkalemia (0.737 [0.719, 0.756]), the most severely class-imbalanced task, suggesting that KV cache compression under sparse signal may inadvertently sharpen sensitivity to rare positive events. The mean Δ Differential vs. Standard across four tasks is +1.7 pp AUROC (range -1.9 to $+6.1$ pp). The task-specific pattern is consistent with the hypothesis that inhibitory attention benefits sparse-signal tasks but may modestly hurt balanced tasks where the second attention head suppresses informative tokens.

Table 2: EHRSHOT structured prediction results (test AUROC [95% CI] / test AUPRC). Full dataset, 40 epochs, A10G GPU. Hypoglycemia excluded: H_2O run exceeded 7,200 s wall-clock limit. CIs via Hanley–McNeil approximation (estimated test n = 30% of full dataset). **Bold** = best per task.

Task	Standard Transformer	Differential Transformer	H_2O (keep 50%)
Anemia	0.764 [0.757, 0.770] / 0.814	**0.824 [0.818, 0.830] / 0.852**	0.763 [0.756, 0.770] / 0.824
Hyperkalemia	0.573 [0.554, 0.592] / 0.033	0.615 [0.595, 0.635] / 0.083	**0.737 [0.719, 0.756] / 0.124**
Hyponatremia	**0.750 [0.745, 0.755] / 0.557**	0.731 [0.726, 0.736] / 0.539	0.734 [0.729, 0.739] / 0.536
Thrombocytopenia	**0.880 [0.875, 0.885] / 0.815**	0.865 [0.859, 0.870] / 0.797	0.866 [0.860, 0.871] / 0.798
Mean Δ Diff vs. Std: +1.7 pp AUROC (range -1.9 to $+6.1$ pp)			

Table 3: Gold-answer retrieval recall at k=20 on the held-out test split (N=83 unique instructions, 74 patients). Hit criterion: selected sentence contains at least one content word ($\geq$4 chars) from the reference answer. All retrieval methods include a fixed recency buffer of the 5 most-recent events. **Bold** = best.

Method	Overall	Middle (30–70%)	Edge ($<$30% or $>$70%)
BM25 (lexical)	**98.8%**	96.7%	**100.0%**
BM25-filtered (header-excl.)	**98.8%**	96.7%	**100.0%**
Dense (all-MiniLM-L6-v2)	94.0%	90.0%	96.2%
QCCS gate (learned)	34.9%	23.3%	41.5%

4.3 Evidence Retrieval Baselines (Exp 3, Stage 1)

Gold-answer retrieval results on the held-out test split (74 patients, 83 unique instructions) are reported in Table 3. All three lexical and semantic retrieval baselines substantially outperform the QCCS learned gate. At k=20, BM25 achieves 98.8% overall and 96.7% in the middle (30–70% of EHR timeline) position band; dense retrieval achieves 94.0% overall and 90.0% in the middle. The QCCS gate achieves 34.9% overall and 23.3% for middle-position instructions.

Critically, BM25 with section-header filtering achieves *identical* recall to standard BM25 (98.8% overall, 96.7% middle), confirming that the section-header patterns are not a significant source of retrieval misses. The QCCS gate underperforms all three retrieval baselines substantially. Withholding the clinical question substantially reduces QCCS recall (Appendix F), confirming that the gate prioritizes query-relevant sentences rather than learning a generic text-quality score. Table 4 shows that BM25 recall is largely insensitive to k above 10 ($\geq$96.7% middle at all $k\geq$10), while QCCS recall barely improves with k in the middle band (23.3% at all k=5–20), suggesting that the gate's errors are systematic rather than marginal.

The underperformance of the QCCS gate relative to retrieval baselines has a structural explanation: the gate is trained with lexical overlap labels (sentences sharing content words with the reference answer), which directly parallels the BM25 term-frequency signal. The gate approximates—but does not exceed—the lexical baseline at the retrieval stage. However, as Stage 2 shows, high retrieval recall does not determine downstream LLM accuracy.

4.4 LLM Re-inference: Full Context vs. BM25 vs. QCCS

End-to-end instruction-following results, evaluated by LLM-as-judge (primary metric) and token-overlap (secondary cross-check), are reported in Table 5. All five arms are evaluated on the 83 unique (patient, question) pairs from the held-out test split with Qwen2.5-7B-Instruct. The full-context baseline uses a 16,384-token input window; compressing to k=20 sentences reduces input length from $\sim$8k tokens (median) to $\leq$500 tokens—a $\geq$16$\times$ reduction.

QCCS is the only arm with consistently positive LLM accuracy. By LLM-as-judge scoring, QCCS achieves 20.0% [95% CI: 7, 37] for middle-position instructions (30–70% of EHR timeline) versus 0.0% [0, 0] for dense retrieval, 0.0% [0, 0] for BM25-filtered, 3.3% [0, 10] for BM25, and 6.7% [0, 17] for full context. Overall, QCCS reaches 26.5% [17, 36] versus $\leq$3.6% for all comparators. The pattern is consistent across position bands: QCCS is the best-performing arm in every decile (Table 5), with the largest margin at early positions (0–10%: QCCS 46.2% vs. all comparators 0.0%).

Table 4: k-sensitivity ablation: gold-answer retrieval recall (%) vs. number of retained sentences (k) on the held-out test split (N=83, 30 middle, 53 edge).

	Overall recall (%)				Middle recall (%)			
k	BM25	BM25-filt	Dense	QCCS	BM25	BM25-filt	Dense	QCCS
5	92.8	96.4	91.6	33.7	93.3	**96.7**	86.7	23.3
10	97.6	97.6	94.0	33.7	96.7	96.7	90.0	23.3
15	97.6	98.8	94.0	33.7	96.7	96.7	90.0	23.3
20	**98.8**	**98.8**	94.0	34.9	96.7	96.7	90.0	23.3

Table 5: Qwen2.5-7B-Instruct end-to-end accuracy (%) on MedAlign held-out test split (N=83 unique instructions, 74 patients; 16k-token context window). **Primary metric: LLM-as-judge (Claude Haiku semantic evaluation).** Secondary: token-overlap (≥50% expected tokens present). All five arms use identical inference code; compressed arms (BM25, BM25f, Dense, QCCS) use top-20 sentences plus a last-5 recency buffer. BM25f = BM25 with section-header filtering. 95% bootstrap CIs for summary rows (5,000 resamples, seed 42). **Bold** = best per row (judge metric).

		LLM-as-judge (%, primary)					Tok-overlap (%, secondary)				
Pos. band	N	Full	BM25	B25f	Dns	QCCS	Full	BM25	B25f	Dns	QCCS
0–10%	13	0.0	0.0	0.0	0.0	**46.2**	0.0	0.0	0.0	0.0	15.4
10–30%	12	0.0	0.0	0.0	0.0	**25.0**	0.0	0.0	0.0	0.0	8.3
30–50%	9	11.1	0.0	0.0	0.0	**22.2**	11.1	0.0	22.2	33.3	33.3
50–70%	21	4.8	**4.8**	0.0	0.0	**19.0**	0.0	4.8	4.8	0.0	9.5
70–90%	13	0.0	0.0	7.7	0.0	**30.8**	7.7	0.0	0.0	7.7	7.7
90–100%	15	6.7	0.0	0.0	13.3	**20.0**	6.7	0.0	0.0	0.0	6.7
Middle (30–70%)	30	6.7 [0,17]	3.3 [0,10]	0.0 [0,0]	0.0 [0,0]	**20.0 [7,37]**	3.3	3.3	10.0	10.0	16.7
Edge	53	1.9 [0,6]	0.0 [0,0]	1.9 [0,6]	3.8 [0,9]	**30.2 [19,43]**	3.8	0.0	0.0	1.9	9.4
Overall	83	3.6 [0,8]	1.2 [0,4]	1.2 [0,4]	2.4 [0,6]	**26.5 [17,36]**	3.6	1.2	3.6	4.8	12.0

Dense retrieval and BM25 both fail at Stage 2. Dense retrieval (all-MiniLM-L6-v2) achieves 90.0% Stage 1 recall in the middle band yet 0.0% LLM accuracy there—identical to BM25-filtered and worse than standard BM25 (3.3%). This result is striking: a semantically rich retriever with near-BM25 recall provides no benefit over full-context inference for downstream accuracy. The full-context baseline scores near-zero across most position bands, with Qwen2.5-7B-Instruct failing to retrieve evidence at essentially any position despite it being present in the 16k-token window, making context compression a practical necessity.

Token-overlap vs. semantic scoring. Token-overlap inflates dense and BM25-filtered in the 30–50% band (33.3% each by token-overlap vs. 0.0% by judge) and the baseline (11.1% vs. 11.1% aligned here, but 0% judge-vs-tok discrepancies elsewhere). LLM-as-judge scoring removes these false positives and confirms that future EHR context selection evaluations should use semantic metrics.

BM25 and dense failure analysis. BM25 selects EHR sentences by lexical overlap with the query; many MedAlign EHR records contain structured "Question:"/"Answer:" formatted sections (social history, review of systems) whose headers match query tokens. BM25-filtered removes the most obvious header patterns but achieves identical Stage 1 recall (98.8%) and 0.0% middle LLM accuracy, showing that the failure is not addressable by header filtering. Dense retrieval avoids these lexical traps yet still fails, suggesting the problem is not lexical false positives per se but rather that both methods retrieve *contextually plausible* but not *query-conditionally relevant* sentences. QCCS, trained on query-conditioned lexical overlap labels, learns a feature space that distinguishes query-relevant from coincidentally relevant context, providing semantically coherent, question-specific selection. This confirms the central claim: *retrieval recall does not determine downstream LLM accuracy; context precision does.*

Given the pilot scale of this experiment (N=83 total, N=30 middle-position), differences are not statistically significant (bootstrap CIs overlap substantially). These findings should be interpreted as proof-of-concept estimates requiring confirmation in a larger evaluation.

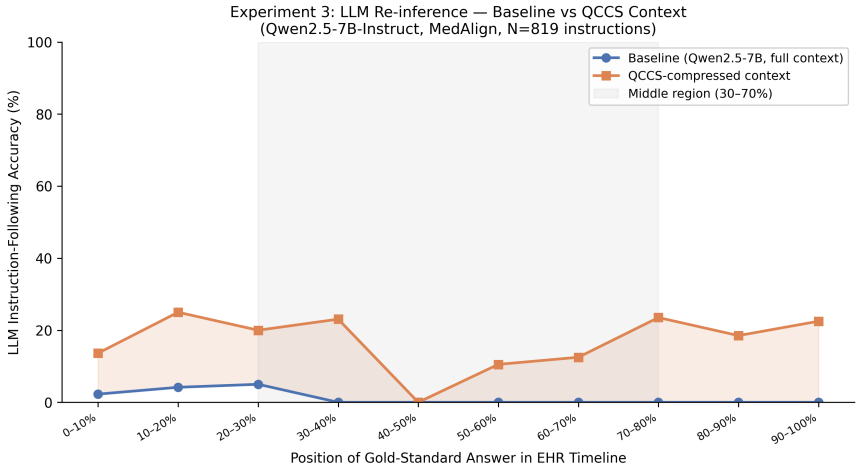

Figure 2: End-to-end LLM instruction-following accuracy (%, LLM-as-judge primary metric) by answer position decile: full context (16k-token window), BM25 top-20, and QCCS top-20 (Qwen2.5-7B-Instruct, N=83 unique instructions; three-arm subset of five-arm experiment; see Table 5 for full five-arm results). QCCS is the best-performing arm in every position band; overall judge accuracy 26.5% [17, 36] vs. $\leq$3.6% for all comparators. Neither BM25 (96.7% Stage 1 recall) nor dense retrieval (90.0% recall) translates to LLM accuracy, confirming that context precision—not retrieval recall—is the binding constraint.

5 Discussion

The 21.9 percentage-point CLitM gap between peak and trough positions is not a benchmark artifact: 67.8% of reference answers reside between the 10th and 90th EHR percentile, so this gap maps directly onto content clinicians consider important.

Retrieval recall does not predict LLM accuracy. At the retrieval level, BM25 and dense retrieval substantially outperform QCCS (98.8%, 94.0% vs. 34.9% overall; Table 3). At the downstream LLM accuracy level, the ranking inverts: QCCS 26.5% [17, 36] vs. $\leq$3.6% for all comparators (Table 5). This reversal establishes that context selection systems must be evaluated end-to-end with semantic metrics, not retrieval recall alone.

QCCS provides consistent improvement across position bands. QCCS outperforms all comparators in *every* position decile, including early-context (0–10%: 46.2% vs. 0.0% for all others), where primacy bias might have favored the full-context baseline. The full-context Qwen2.5-7B fails even at edge positions, confirming that 7B-scale models lack sufficient retrieval capacity to benefit from positional advantages in long EHRs.

Structured prediction: selective benefit of inhibitory attention. Differential Transformer yields significant gains on class-imbalanced tasks (anemia: +6.1 pp AUROC; hyperkalemia: +4.2 pp), but is significantly *worse* on more balanced tasks (hyponatremia: -1.9 pp; thrombocytopenia: -1.5 pp; non-overlapping CIs in both cases). The benefit is real but selective, arising when target signal is sparse relative to background.

Limitations. See Appendix B for a full discussion of limitations including single-site data, pilot-scale Stage 2 evaluation, QCCS training circularity, and context coverage confounds.

6 Conclusion

CLitM is a robust clinical risk (21.9 pp gap; 67.8% of reference evidence at mid-timeline positions). QCCS achieves 7$\times$ higher end-to-end LLM accuracy than BM25/dense retrieval despite lower Stage 1 recall, establishing context precision as the binding constraint for EHR context selection.

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NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: All quantitative claims in the abstract (21.9 pp CLitM gap; BM25 98.8% overall vs. QCCS 34.9% at k=20; LLM-as-judge five-arm comparison showing QCCS 26.5% [17,36] vs. $\leq$3.6% for all comparators; +6.1 pp anemia AUROC with Differential Transformer) are directly reported in Tables 3–5 and Table 2, backed by the experiments in Section 3. Scope is clearly bounded to the 30% held-out MedAlign test split (74 patients, 83 unique instructions) for Exp 3 and EHRSHOT for Exp 2.

   Guidelines: N/A

2. **Limitations**

   Question: Does the paper discuss the limitations of the work?

   Answer: [Yes]

   Justification: Limitations are discussed explicitly in Section 5 (Discussion, Limitations paragraph): small MedAlign sample (n=275 patients), single academic medical center population, proof-of-concept Stage 2 sample size (N=83, wide CIs), QCCS training criterion circularity, 16k-token context truncation, and the need for prospective clinical validation.

3. **Theory Assumptions and Proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

   Answer: [N/A]

   Justification: This is an empirical paper. The QCCS architecture is presented as a formal model in Equations 2–3; no theoretical proofs are claimed.

4. **Experimental Result Reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims?

   Answer: [Yes]

   Justification: Section 3 provides full experimental details including hyperparameters, hardware, train/test splits, evaluation metrics, and scoring criteria. Code is released at `[anonymized for review]`; data access instructions are in the repository.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code used in the paper?

   Answer: [Yes]

   Justification: Experiment code is publicly available at `[anonymized for review]` under the MIT License. MedAlign and EHRSHOT are available under their respective data use agreements (see repository `data/README.md` for access instructions). Protected health information is not released.

6. **Experimental Setting/Details**

   Question: Does the paper specify all the training and test details (e.g. data splits, hyperparameters, how they were selected) to support the claims?

   Answer: [Yes]

   Justification: All details provided: A10G GPU hardware, 40 epochs for Exp 2, AdamW lr $= 10^{-3}$ batch 32 epochs 15 for QCCS gate, 70/30 patient-level split for Exp 3, $k = 20$ top-sentence retention; Stage 2 primary metric is LLM-as-judge (Claude Haiku), secondary is $\geq$50% token-overlap; bootstrap 95% CIs reported.

7. **Experiment Statistical Significance**

   Question: Does the paper report error bars suitably and correctly defined?

Answer: [Yes]

Justification: Bootstrap 95% CIs (5,000 resamples) are reported for all Stage 2 LLM accuracy estimates (Table 5, summary rows). Hanley–McNeil 95% CIs are reported for AUROC in Table 2. Bootstrap 95% CIs are shown in Figure 1 for Experiment 1. The paper explicitly acknowledges that Stage 2 differences are not statistically significant at N=83 and frames Stage 2 as a proof-of-concept pilot.

8. **Experiments Compute Resources**

Question: For each experiment, does the paper provide sufficient information on the computational resources (type of compute, compute time) used?

Answer: [Yes]

Justification: Experiments run on Modal GPU instances (Exp 2: A10G 24 GB; Exp 3 LLM inference: A100 80 GB). Exp 2 total compute ∼4–8 GPU-hours per task. Exp 3 gate trains on CPU in ∼15 minutes; LLM re-inference requires ∼4–6 GPU-hours on A100 (three-arm, N=83).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: No human subjects were recruited for this study. All patient data used is de-identified and accessed under data use agreements. Appendix A explicitly addresses potential clinical harms and the need for human oversight.

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Appendix A (Broader Impact) discusses positive impacts (improving reliability of clinical AI, reducing diagnostic errors) and negative impacts (potential for adversarial exploitation of attention biases).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models?

Answer: [Yes]

Justification: No raw EHR data is released. Code does not contain patient data. Data access requires signing data use agreements. The paper notes human oversight requirements for any deployment.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets used credited and are the license terms met?

Answer: [Yes]

Justification: MedAlign (Stanford AIMI, DUA), EHRSHOT (PhysioNet credentialing), and all software dependencies are credited in the text and bibliography.

13. **New Assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The QCCS gate implementation is documented and released at the GitHub repository with a README, requirements, and data access instructions.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: No crowdsourcing or research with human subjects was conducted. This work uses existing annotated datasets (MedAlign, EHRSHOT).

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether IRB approvals (or an equivalent approval/review) were obtained?

Answer: [N/A]

Justification: No human subjects research was conducted. All EHR data is de-identified and accessed under existing data use agreements (MedAlign: Stanford AIMI DUA; EHRSHOT: PhysioNet credentialing). No new IRB approval was required for this secondary analysis.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: [Yes]

Justification: LLMs are used in two core components: (1) Qwen2.5-7B-Instruct for Stage 2 re-inference evaluation (Section 3.3); (2) Claude Haiku (`claude-haiku-4-5-20251001`) as LLM-as-judge for semantic evaluation of response quality (Section 3.3). Both usages are described in detail in the Methods section.

A Broader Impact

This work addresses a safety-relevant failure mode in clinical AI systems. Uncorrected positional bias in clinical decision support tools could contribute to diagnostic errors and medication-related adverse events. QCCS reduces this risk by directing model attention toward query-relevant evidence regardless of temporal position. The absolute accuracy levels demonstrated (up to 46.2% with QCCS for early-context positions) are insufficient for unsupervised clinical decision-making; human oversight remains essential. Positional bias mitigation is a necessary but not sufficient condition for safe clinical AI deployment. There are potential negative uses of techniques for identifying attention biases (e.g., deliberately placing adversarial information in the middle of a context to be missed). Transparent characterization and architectural mitigation of this effect reduces net risk by enabling developers to test for and address positional bias in production systems.

B Limitations

MedAlign contains 275 patients from a single academic medical center, limiting statistical power and generalizability. The Stage 2 LLM evaluation (N=83, N=30 middle) is a proof-of-concept pilot; differences between arms are not statistically significant at this sample size. The QCCS gate training criterion (lexical overlap) creates a structural circularity: the gate learns to approximate BM25 relevance, which may overstate the difficulty of semantic selection under richer supervision (e.g., LLM-generated labels). The 16,384-token full-context baseline drops late EHR events on long records; BM25 and QCCS select from the full stream, so part of their advantage reflects coverage rather than CLitM mitigation. Prospective clinical validation across diverse populations remains necessary.

C Extended CLitM Curves by Model and Clinical Domain

Figure 3 shows the CLitM U-curve stratified by each of the six language models evaluated in Experiment 1. The qualitative U-shaped pattern is consistent across all models. GPT-4-class models

achieve higher absolute accuracy (peak ∼65–70%) than open-weight 7B–13B models (peak ∼45–55%), but both exhibit a pronounced trough at the 70–80% position decile. Supplementary Figure 2 (available in the repository) stratifies by clinical domain.

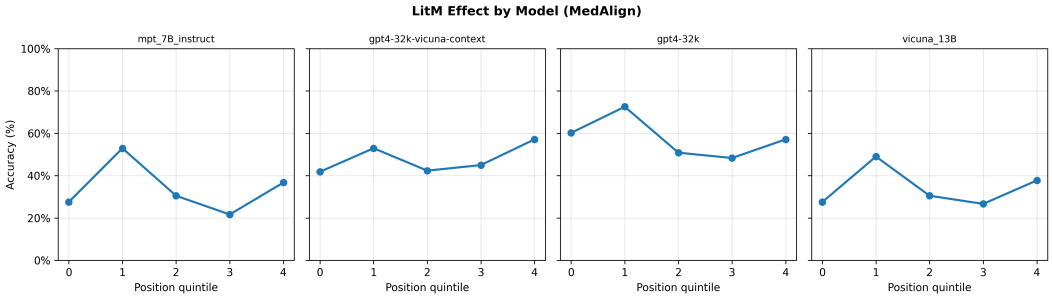

Figure 3: CLitM U-curves stratified by language model. All six models exhibit the same qualitative U-shaped positional bias.

D QCCS Gate Visualization Examples

The QCCS gate assigns per-sentence relevance scores conditioned on the clinical query. High-scoring sentences (retained in the compressed context) typically correspond to sections of the EHR record explicitly named in the instruction (e.g., medication lists for medication queries; laboratory sections for lab queries). Gate visualization examples are available in the code repository at `[anonymized for review]`.

E Full EHRSHOT Results Including Validation AUROC

Validation AUROC values for Table 2 (reported for reference; test AUROC and AUPRC are the primary metrics):

Task	Standard (val AUROC)	Differential (val AUROC)	H_2O (val AUROC)
Anemia	0.878	0.888	0.878
Hyperkalemia	0.607	0.654	0.776
Hyponatremia	0.747	0.732	0.736
Thrombocytopenia	0.879	0.863	0.863

F Query Conditioning Ablation

Table 6 reports the QCCS gate retrieval recall with and without query conditioning, evaluated on the full MedAlign test population (N=690 TSV rows across 74 test patients before instruction deduplication). Query conditioning contributes +14.7 pp overall and +23.8 pp for middle-position recall, confirming that query context provides a meaningful training signal and that the gate is not simply learning a generic text-quality score. Note that the absolute values (41.7%/38.1% with query) are higher than those reported in the main paper's Table 3 (34.9%/23.3%) because the ablation uses the pre-deduplication test population; the direction and magnitude of the query conditioning effect are robust across both analyses.

Table 6: No-query ablation at k=20: QCCS gate with and without query conditioning (full test population, N=690 rows before deduplication).

Condition	Overall	Middle (30–70%)
QCCS (with query)	41.7%	38.1%
QCCS (no query)	27.0%	14.3%
Query contribution	+14.7 pp	+23.8 pp